

Angles, polygons, congruence and vectors - Set 2

TMUA practice questions by topic. Specification M5.1-M5.6, M5.10, M5.14, M5.18, M5.19. 4 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 In triangle OAB , $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$. M is the midpoint of AB and P is the point on OM with $OP : PM = 2 : 1$. Express $\vec{AP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

- A $\frac{2}{3}(\mathbf{a} + \mathbf{b})$
B $\frac{2}{3}\mathbf{b} - \frac{1}{3}\mathbf{a}$
C $\frac{1}{3}(\mathbf{b} - \mathbf{a})$
D $\frac{2}{3}\mathbf{b} - \frac{1}{3}\mathbf{a}$
E $\frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$
F $\frac{1}{2}\mathbf{b} - \frac{1}{2}\mathbf{a}$

- 2 A ship sails from P on a bearing of 060° for 10 km to Q , then on a bearing of 150° for 10 km to R . What is the bearing of P from R ?

- A 255°
B 195°
C 315°
D 285°
E 075°
F 105°

- 3 $\vec{AB} = 3\mathbf{p} + 2\mathbf{q}$ and $\vec{BC} = k\mathbf{p} - 4\mathbf{q}$, where $\mathbf{p}$ and $\mathbf{q}$ are not parallel. Given that A , B and C are collinear, find k .

- A 2
B 6
C -3
D $\frac{3}{2}$
E -6
F -4

- 4 A regular hexagon and a regular octagon share a common side. What is the size of the angle between their two other sides that meet at one end of the common side (the angle outside both polygons)?
- A 120°
 - B 95°
 - C 105°
 - D 135°
 - E 115°
 - F 75°